Lab - V

Section: F

Name: Neeraj R Rugi

SRN: PES2UG24CS311

1. D Flip-Flop:

Code:

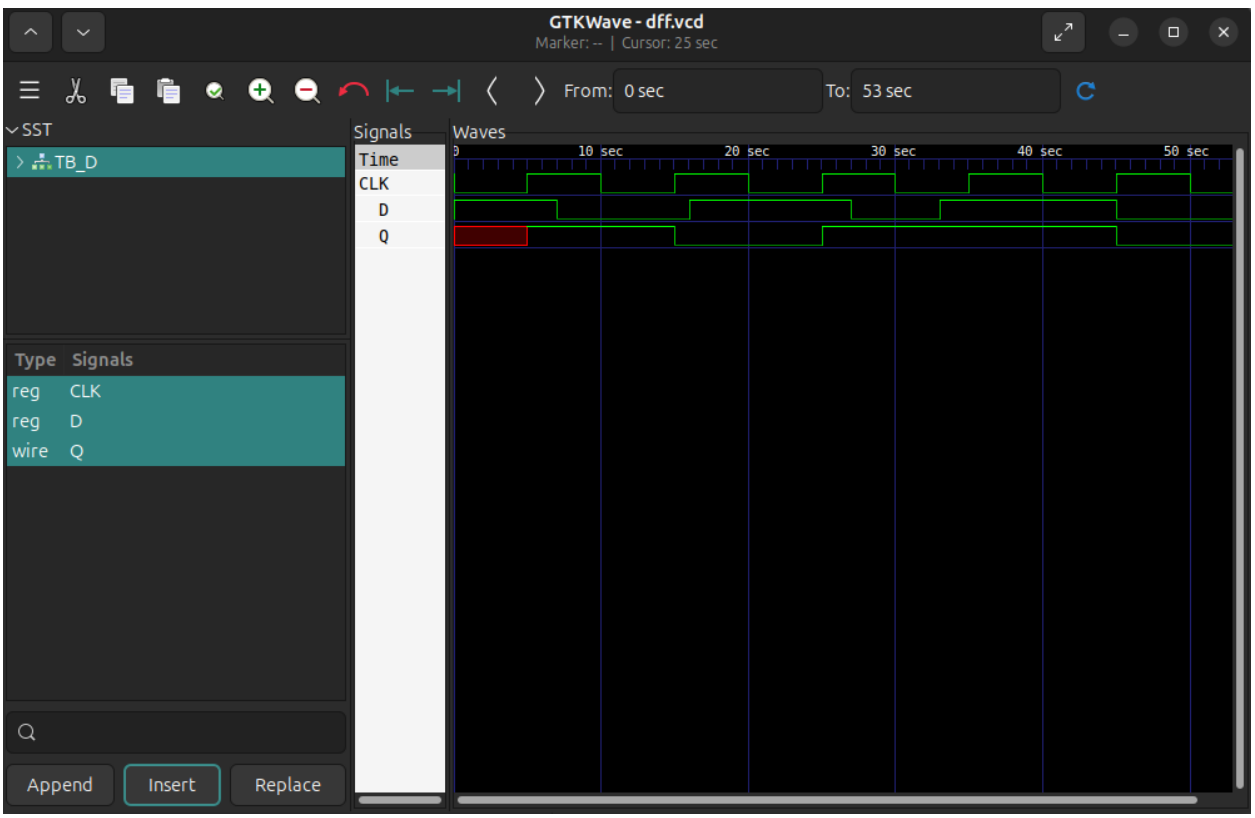
```
module dff (input wire D, input wire clk, output reg Q);
  always @(posedge clk)
    begin

      Q <= D; // Sample D on rising edge of clock
    end
endmodule
```

```verilog
module TB_D;
reg D;
reg CLK;
wire Q;
dff newD (.D(D), .clk(CLK),.Q(Q));
 // clock
initial begin
  CLK = 1'b0;
  forever #5 CLK = ~CLK;
end
initial
begin
// waveform dump
$dumpfile("dff.vcd");
$dumpvars(0, TB_D);
end
initial
begin
    $monitor($time, " CLK=%b, D=%b, Q=%b", CLK, D, Q);
    D = 1'b1; #7;
    D = 1'b0; #9;
    D = 1'b1; #11;
    D = 1'b0; #6;
    D = 1'b1; #12;
    D = 1'b0; #8;
    $finish;
end
initial
begin
    #200 $finish;
end
endmodule
```

VVP:

```
VCD info: dumpfile dff.vcd opened for output.
                   0 CLK=0, D=1, Q=x
                   5 CLK=1, D=1, Q=1
                   7 CLK=1, D=0, Q=1
                  10 CLK=0, D=0, Q=1
                  15 CLK=1, D=0, Q=0
                  16 CLK=1, D=1, Q=0
                  20 CLK=0, D=1, Q=0
                  25 CLK=1, D=1, Q=1
                  27 CLK=1, D=0, Q=1
                  30 CLK=0, D=0, Q=1
                  33 CLK=0, D=1, Q=1
                  35 CLK=1, D=1, Q=1
                  40 CLK=0, D=1, Q=1
                  45 CLK=1, D=0, Q=0
                  50 CLK=0, D=0, Q=0
dff_tb.v:26: $finish called at 53 (1s)
```

GTKWave:



2. T Flip-Flop:

Code:

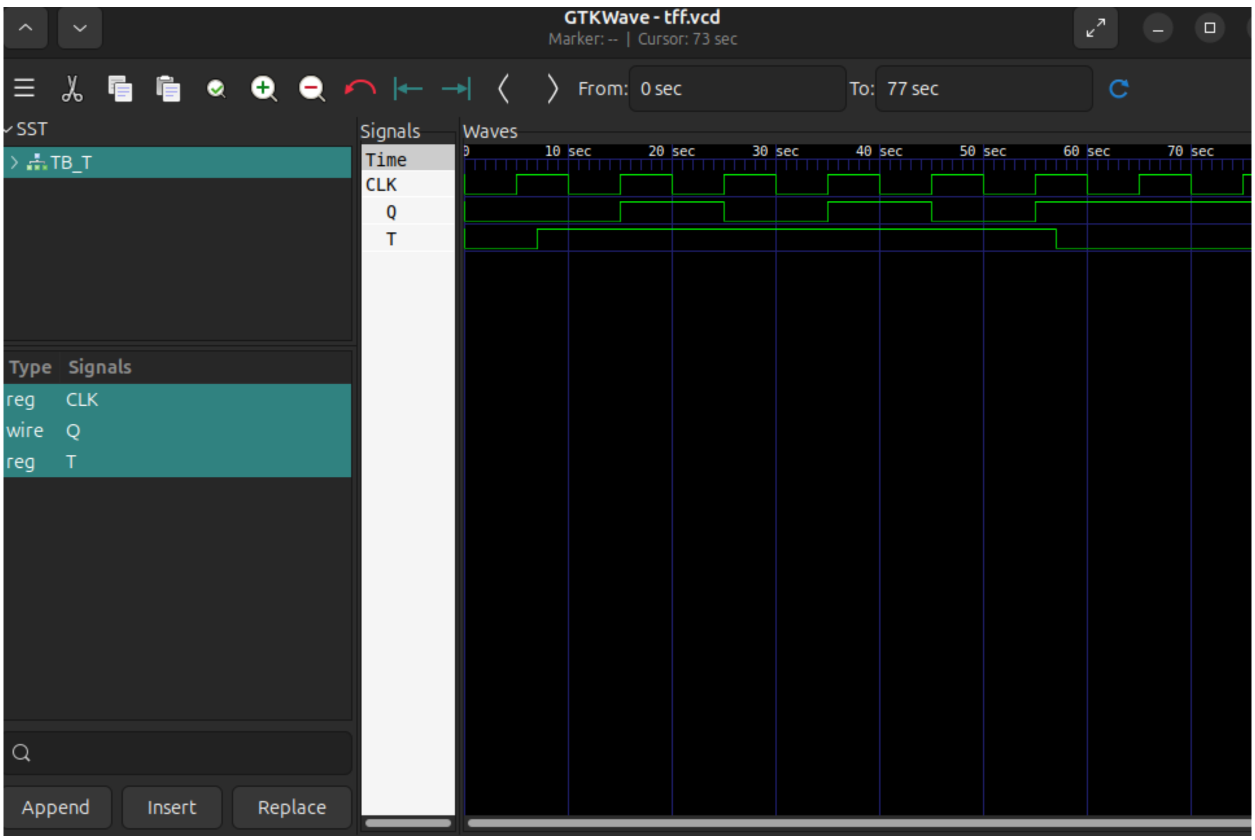
```
module tff (input wire T, input wire clk, output reg Q);
initial Q = 0;
always @(posedge clk )
        begin
    if (T)    Q <= ~Q;    // toggle
    else    Q <= Q;    // hold
  end
endmodule
```

```
module TB_T;
reg T;
reg CLK;
wire Q;
 // Instantiate T flip-flop
 tff newT (.T(T), .clk(CLK), .Q(Q));
// Clock generation: 10 time units period
initial
begin
    CLK = 1'b0;
forever #5 CLK = ~CLK;
end
// Waveform dump
 initial
begin
 $dumpfile("tff.vcd");
 $dumpvars(0, TB_T);
end
initial
begin
$monitor("Time=%0t T=%b | Q=%b", $time,T, Q);
  end
initial
begin
   T = 1'b0;
   #7 T = 1'b1;
   #50 T = 1'b0;
   #20 $finish;
end

initial
begin
   #200 $finish;
end
endmodule
```

VVP:

```
VCD info: dumpfile tff.vcd opened for output.
Time=0 T=0 | Q=0
Time=7 T=1 | Q=0
Time=15 T=1 | Q=1
Time=25 T=1 | Q=0
Time=35 T=1 | Q=1
Time=45 T=1 | Q=0
Time=55 T=1 | Q=1
Time=57 T=0 | Q=1
tff_tb.v:30: $finish called at 77 (1s)
```

GTKWave:



3. Jack-Killby(JK) Flip-Flop

Code:

```
module jkff (input wire j, input wire k,input wire clk,output reg q );
initial q = 0;
    always @(posedge clk)
    begin
    begin case ({j, k})
        2'b00: q <= q;          // No change
        2'b01: q <= 1'b0;       // Reset
        2'b10: q <= 1'b1;       // Set
        2'b11: q <= ~q;         // Toggle
    endcase
    end
    end
endmodule
```

```
module tb_jk_ff;
reg j, k, clk;
wire q;
 // Instantiate JK flip-flop
jkff jk1 (.j(j),.k(k),.clk(clk),.q(q));
// Generate clock
initial begin
    clk = 0;
forever #5 clk = ~clk;
end
initial
begin
// waveform dump
$dumpfile("jkff.vcd");
$dumpvars(0, tb_jk_ff);
end
// Test sequence
initial begin        // Monitor outputs
    $monitor("Time=%0t | J=%b K=%b | Q=%b", $time, j, k, q);
    j = 0; k = 0;
    #10 j = 0; k = 0;   // Hold
    #10 j = 0; k = 1;   // Reset
    #10 j = 1; k = 0;   // Set
    #10 j = 1; k = 1;   // Toggle
    #10 j = 1; k = 1;   // Toggle again
    #10 j = 0; k = 1;   // Reset
    #10 j = 1; k = 0;   // Set
    #20 $finish;
end
endmodule
```

VVP:

```
VCD info: dumpfile jkff.vcd opened for output.
Time=0 | J=0 K=0 | Q=0
Time=20 | J=0 K=1 | Q=0
Time=30 | J=1 K=0 | Q=0
Time=35 | J=1 K=0 | Q=1
Time=40 | J=1 K=1 | Q=1
Time=45 | J=1 K=1 | Q=0
Time=55 | J=1 K=1 | Q=1
Time=60 | J=0 K=1 | Q=1
Time=65 | J=0 K=1 | Q=0
Time=70 | J=1 K=0 | Q=0
Time=75 | J=1 K=0 | Q=1
jkff_tb.v:28: $finish called at 90 (1s)
```

GTKWave: